

50.007 Machine Learning

Course Project Brief

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Agenda

- Feature Engineering and TF/IDF
- Project Overview
- Project Tasks
- Intro to Kaggle & Administrative

Feature Engineering

- Feature engineering is the process of selecting, manipulating, and transforming raw data into features that can be used in supervised learning.
 - Represent your model's x with *features*

Diagram illustrating a dataset structure for supervised learning. The table shows a record/instance x (rows) and its corresponding label y (columns).

label y	attributes/ features				
Survival	Age	Sex	Fare	Cabin	...
Yes	67	F	Normal	003	...
No	32	M	Premium	021	...
...	...	...	...	...	...

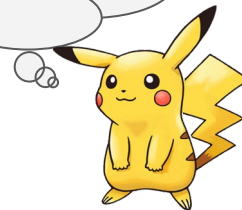
How do we represent text as features?

- How do you represent the following texts as features?
 - *Lyrics 1: baby, Im dancing in the dark*
 - *Lyrics 2: baby, baby, baby, oh*
 - *Lyrics 3: baby shark, do do, do do do do*

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How about one-hot encoding?



One-Hot Encoding

- How do you represent the following texts as features?
 - *Lyrics 1: baby, Im dancing in the dark*
 - *Lyrics 2: baby, baby, baby, oh*
 - *Lyrics 3: baby shark, do do, do do do do*
- Step 1: Identify all unique words in corpus

<i>baby</i>		<i>the</i>
	<i>do</i>	
<i>im</i>		<i>dark</i>
<i>dancing</i>	<i>oh</i>	
<i>in</i>		<i>shark</i>

One-Hot Encoding

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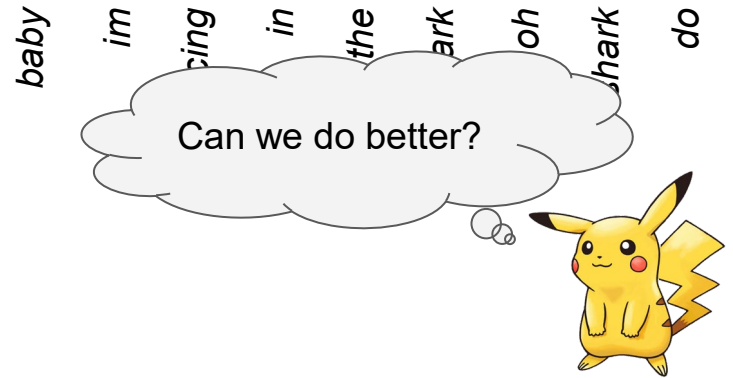
- Step 1: Identify all unique words in corpus
- Step 2: Represent each lyrics with a binary vector indicating if the word is present

1	1	1	1	1	1	0	0	0
1	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	1	1
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

One-Hot Encoding

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- Step 2: Represent each lyrics with a binary vector indicating if the word is present

1	1	1	1	1	1	0	0	0
1	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	1	1



Term Frequency

- How do you represent the following texts as features?

- Lyrics 1: *baby, Im dancing in the dark*
- Lyrics 2: *baby, baby, baby, oh*
- Lyrics 3: *baby shark, do do, do do do do*

- Term Frequency (TF) tells you how “important” is the term in the document

<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

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- How do you represent the following texts as features?

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$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

Number of times term t occur in document d

Total number of terms t' in document d

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- Term Frequency (TF) tells you how “important” is the term in the document

0.17								
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

Number of times term t occur in document d

Total number of terms t' in document d

$$tf(baby, lyric1) = \frac{1}{6} = 0.17$$

Term Frequency

- How do you represent the following texts as features?
 - Lyrics 1: *baby, Im dancing in the dark*
 - Lyrics 2: *baby, baby, baby, oh*
 - Lyrics 3: *baby shark, do do, do do do do*
- Term Frequency (TF) tells you how “important” is the term in the document

0.17								
0.75								
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

Number of times term t occur in document d

Total number of terms t' in document d

$$tf(baby, lyric2) = \frac{3}{4} = 0.75$$

Term Frequency

- How do you represent the following texts as features?
 - Lyrics 1: baby, Im dancing in the dark*
 - Lyrics 2: baby, baby, baby, oh*
 - Lyrics 3: baby shark, do do, do do do do*
- Term Frequency (TF) tells you how “important” is the term in the document

0.17	0.17	0.17	0.17	0.17	0.17	0	0	0
0.75	0	0	0	0	0	0.25	0	0
0.13	0	0	0	0	0	0	0.13	0.75
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

Number of times term t occur in document d

Total number of terms t' in document d

Term Frequency

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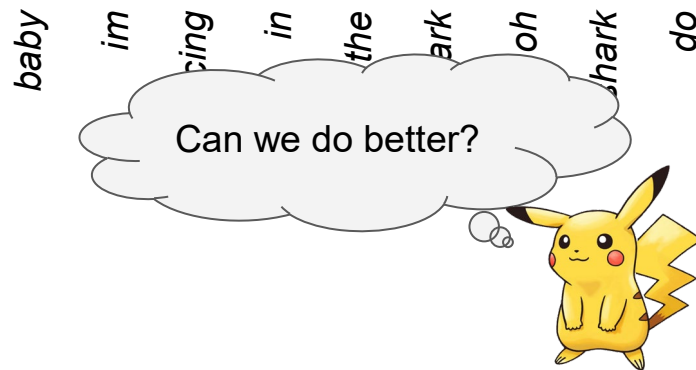
- Term Frequency (TF) tells you how “important” is the term in the document

$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

Number of times term t occur in document d

Total number of terms t' in document d

0.17	0.17	0.17	0.17	0.17	0.17	0	0	0
0.75	0	0	0	0	0	0.25	0	0
0.13	0	0	0	0	0	0	0.13	0.75



Inverse Document Frequency

- How do you represent the following texts as features?

- Lyrics 1: *baby, Im dancing in the dark*
- Lyrics 2: *baby, baby, baby, oh*
- Lyrics 3: *baby shark, do do, do do do do*

- Inverse Document Frequency (IDF) measures how much information the word provides

<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$\text{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|}$$

Inverse Document Frequency

- How do you represent the following texts as features?
 - Lyrics 1: baby, Im dancing in the dark
 - Lyrics 2: baby, baby, baby, oh
 - Lyrics 3: baby shark, do do, do do do do
- Inverse Document Frequency (IDF) measures how much information the word provides

baby	im	dancing	in	the	dark	oh	shark	do

$$\text{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|}$$

N Total number documents in dataset
 $|\{d \in D : t \in d\}|$ Number of document d with term t

Inverse Document Frequency

- How do you represent the following texts as features?

- Lyrics 1: *baby, Im dancing in the dark*
- Lyrics 2: *baby, baby, baby, oh*
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- Inverse Document Frequency (IDF) measures how much information the word provides

0								
0								
0								
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$\text{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|}$$

N Total number documents in dataset
 $|\{d \in D : t \in d\}|$ Number of document d with term t

$$\text{idf}(\text{baby}, D) = \log \frac{3}{3} = 0$$

Inverse Document Frequency

- How do you represent the following texts as features?

- Lyrics 1: *baby, Im dancing in the dark*
- Lyrics 2: *baby, baby, baby, oh*
- Lyrics 3: *baby shark, do do, do do do do*

- Inverse Document Frequency (IDF) measures how much information the word provides

0							0.48	
0							0.48	
0							0.48	
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$\text{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|}$$

N Total number documents in dataset
 $|\{d \in D : t \in d\}|$ Number of document d with term t

$$\text{idf}(\text{shark}, D) = \log \frac{3}{1} = 0.48$$

Inverse Document Frequency

- How do you represent the following texts as features?
 - Lyrics 1: *baby, Im dancing in the dark*
 - Lyrics 2: *baby, baby, baby, oh*
 - Lyrics 3: *baby shark, do do, do do do do*
- Inverse Document Frequency (IDF) measures how much information the word provides

0	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
0	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
0	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

$$\text{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|}$$

N Total number documents in dataset
 $|\{d \in D : t \in d\}|$ Number of document d with term t

TF-IDF

- How do you represent the following texts as features?

- Lyrics 1: *baby, Im dancing in the dark*
- Lyrics 2: *baby, baby, baby, oh*
- Lyrics 3: *baby shark, do do, do do do do*

0	0.08	0.08	0.08	0.08	0.08	0	0	0
0	0	0	0	0	0	0.12	0	0
0	0	0	0	0	0	0	0.06	0.36
<i>baby</i>	<i>im</i>	<i>dancing</i>	<i>in</i>	<i>the</i>	<i>dark</i>	<i>oh</i>	<i>shark</i>	<i>do</i>

- TF-IDF calculated as:

$$\text{tfidf}(t, d, D) = \text{tf}(t, d) \cdot \text{idf}(t, D)$$

Project Overview

- Objectives:
 1. Consolidate what we have learned in class
 2. Hands-on opportunity - Equip with practical ML skills for your future job and internship!
 3. Practical assessment!
- Check out:

<https://www.kaggle.com/competitions/50-007-machine-learning-may-2026>

Project Overview

As AI generated contents are everywhere, and they will only be harder to differentiate between human and AI generated content.

In this project, we will be looking at **GenAI content detection**. The problem statement is:

“Given a piece of text, classify if this is human-authored text or machine-generated content.”

Dataset

- We sampled a subset of data from the [GenAIDetect Workshop](#) held in COLING 2025.
- The data splits as follows:
 - 20,000 train set
 - 2,000 dev set
 - 5,000 test set
- However, this is only <5% of their dataset, hence you should not be expecting the performance will be great if you run on the full dataset.
- You are encouraged to read about the dataset [here](#). (Section 3.2 English subtask, and Section 5.1)

Wang, Y., Shelmanov, A., Mansurov, J., Tsvigun, A., Mikhailov, V., Xing, R., ... & Nakov, P. (2025, January). GenAI content detection task 1: English and multilingual machine-generated text detection: AI vs. human. In Proceedings of the 1st Workshop on GenAI content detection (GenAIDetect) (pp. 244-261).

Dataset

- In Kaggle, we have provided you the following files:
 - train.csv
 - test.csv
 - train_features.csv
 - test_features.csv
 - sample_submission.csv

}	Original text
}	Preprocessed features
}	Sample submission
- You can preprocess the text by your own, but not necessary.
 - You **MUST** use our preprocessed features for Task 1 and 2.
 - If you chose to preprocess the text for Task 3, you **MUST** share what have you tried as well.
 - **NO** extra marks will be awarded.

Project Tasks

In this project, we have 3 tasks.

1. Implement Logistic Regression from Scratch (5 marks)
2. Implement PCA and KNN (5 marks)
3. Compete in leaderboard* (15 marks)

Finally, your report and presentations** (25 marks)

* Evaluation is based on Macro-F1 score, not accuracy!

** Your learning process is more important than you “implemented” a model that outperforms a LLM.

Task 1

Implement a logistic regression from scratch.

- You are **NOT ALLOWED** to use any predefined logistic regression package for this task, for example sklearn!

In your final submission, please include:

- Implementation code (Jupyter Notebook)
- Prediction by your model on test set
 - Please name it as “LogReg_predictions.csv”

Task 1 Rubrics

Task 1				
Score	0	1-2	3-4	5
Rubrics	No submission or using pre-defined logistic regression module/libraries such as sklearn.	Incorrectly implemented logistic regression algorithm with compilation errors.	Implemented logistic regression algorithm with minor errors. The implemented model is compilable but has issues training the implemented model with the train set.	Perfect implementation of logistic regression algorithm. Successfully trained the model with the train set and achieved comparative performance compared to sklearn logistic regression

Task 2

Implement PCA for feature dimension reduction, and perform classification using KNN model, please set `n_neighbors=2`.

- You can use sklearn library for this task

In your final submission, please include:

- Implementation code
- Report the Macro F1 scores for applying **2000, 1000, 500, and 100 components** on the test set
 - You only need to include this in your final report

Task 2 Rubrics

Task 2				
Score	0	1-2	3-4	5
Rubrics	No submission.	Incorrectly implemented PCA and KNN. Unable to reduce the train set feature dimensions and run KNN on the test sets with reduced components	Correctly implemented PCA and KNN with minor errors. The implemented model is able to run on test sets with reduced components	Perfectly implemented PCA and KNN. The implemented model is able to run on test sets with reduced components and performed detailed analysis of the reduced components

Task 3

Apply any machine learning models and compete in the leaderboard!

- **DO NOT** use any deep learning approach (including LLMs)!

For the top-3 teams on the private leaderboard, bonus marks will be provided*:

- Third-highest score: extra 1 marks
- Second-highest score: extra 2 marks
- Highest score: extra 3 marks

* Capped at 15 marks

Task 3

In your final submission, please include:

- Implementation code for your best performing model
- Prediction by your best performing model on public leaderboard

Task 3 Rubrics

Task 3				
Score*	0	1-5	6-10	11-15
Rubrics	No submission.	Final submission is not able to beat the red line in private leaderboard. The team have explored 1-2 machine learning models.	Final submission is able to beat the red line in private leaderboard. The team have explored 2-3 machine learning models, with proper documentation on the hyperparameter tuning.	Final submission is able to beat the red and blue line in private leaderboard. The team have explored more than 4 machine learning models, with proper documentation on the hyperparameter tuning.

*For Task 3, you will be able to get bonus marks (cap to 15 marks)

Task 4

Submit your final report! You should answer the following questions:

1. Introduce your best performing model on how it works.
2. Discuss how you achieved your best performing model, how you “tune” it. What are the parameters that you have used and the different parameters you have tried to achieve the best results. (Tell us your roadmap)
3. Are there any difficulties you faced when you were tuning the models? If you have and overcome these difficulties, what were the actions taken? If you failed to overcome the difficulties, tell us your reflection.
4. Did you self-learn anything that is beyond the course? If yes, what are they, and tell us more about it.

Task 4 Rubrics

Task 4 (Report)				
Score	0	1-5	6-10	11-15
Rubrics	No submission.	Documentation is inadequate to explain the models explored.	Documentation is adequate to explain the models explored. The documentation answered the questions listed under this task	Documentation is adequate to explain the models explored. The documentation answered the questions listed under this task creatively with interesting insights

Presentation Rubrics

Restricted

Task 4 (Presentation)

Score	0	1-3	4-7	8-10
Rubrics	Students did not attend or did not deliver the presentation.	Presentation lacks organization and clarity. Students provide limited explanation of their algorithm and demonstrate insufficient understanding of key concepts. They are unable to adequately answer questions about the design, implementation, or rationale behind their approach.	Presentation is clear and reasonably well-structured. Students explain the main components of their algorithm and demonstrate a good understanding of their work. They are able to answer most questions related to the methodology, implementation choices, and results.	Presentation is engaging, well-organized, and professionally delivered. Students clearly articulate the motivation, design decisions, and limitations of their algorithm. They demonstrate deep understanding by confidently answering questions and providing thoughtful analysis, critical reflection, or interesting insights beyond the basic requirements.

Restricted

Project Groupings

- Each team should have 5 students. Please register here:
<https://docs.google.com/spreadsheets/d/1N8uECmIKZI4gWaBFIpYta9FZLTeyytIY1F-r7UJhUAM/edit?usp=sharing>
- Each team (at least one member) should register account in Kaggle [here](#).
 - You can use your personal email to join, grading are based on group names.
- Join the competition using this [link](#).

Forming a Group

Click “Team”

Overview Data Code Models Discussion Leaderboard Rules **Team** Submissions Settings

Your Team

Everyone that competes in a Competition does so as a team - even if you're competing by yourself. [Learn more.](#)

General

TEAM NAME

GROUP_NAME_HERE

This name will appear on your team's leaderboard position.

Change
to your
group
name

Forming a Group

Team Members

Your team can have a maximum of 5 members.



Rui Yang Tan (You)

Team Leader

Search for your friends and invite them to join your team

Send Invitation

You can invite 4 more people to join your team. They will need to have joined the competition.



Search for the team or user name

Your friend should receive the invitation here

Received Invitations

You haven't received invitations from other teams.

Finding Information

All instructions are in the “Overview” tab, please read through!

Settings Overview Data Code Models Discussion Leaderboard Rules Team

Project Summary

Large Language Models (LLMs) have demonstrated remarkable success and gained widespread popularity, becoming a key technology behind generative artificial intelligence (GenAI) applications used in everyday life. However, this growing adoption has also raised several concerns, including threats to the integrity of student assignments and the risk of fabricated content misleading individuals. As generative AI continues to evolve, it is becoming increasingly difficult for humans to distinguish between machine-generated and human-written text (Wang et.al, 2025). Feel free to talk to us if you want to know why! :)

In this project your task is to perform a text classification task: "Given a piece of text, classify if this is human-authored text or machine-generated content". You are to apply machine learning approaches to perform this GenAI content detection task. Specifically, you need to perform the following tasks.

Table of Contents

- Overview
- Description
- Evaluation
- Deliverables And D...
- Grading Rubrics

+ Add Section

Tips to do well in course project

1. Read the instructions clearly! Clarify any doubts!
2. Read the grading metrics to know how to score well!
3. **DO NOT** spend too much time on the project (especially Task 3!) - There is no perfection 100% Macro-F1 score (that's overfitting).
 - a. **DO NOT** manually label the test set - WE WILL KNOW!
4. Source out solutions beyond what was taught in class! Understand and be able to explain the solutions clearly!
5. You are encouraged to ask questions, but not the baselines